MIL-L-48766 (AR)
AMENDMENT 2
17 December 1979
SUPERSEDING
AMENDMENT 1
25 September 1978

MILITARY SPECIFICATION

LITHIUM TETRAFLUOROBORATE
(FOR USE IN AMMUNITION)

This Amendment forms a part of Military Specification MIL-L-48766(AR) dated 19 June 1975.

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Delete "(PA)" wherever it appears and substitute "(AR)".

Delete the Preamble and substitute the following:

"This specification approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense".

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Para. 3.1.2. X-ray diffraction. Add "(see 3.1.3)".

Para. 3.1.3. Infrared Spectroscopy. Add the following: "This requirement may be used as an alternate to x-ray diffraction requirement given in 3.1.2."

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Para 3.1.4. Impurities. Delete the last sentence in the paragraph.
Delete Table III in its entirety and substitute the following Table III.

Table III"

" Impurities	Maximum Allowable Concentration, ppm	Test Paragraph
Total Halogens	50	4.4.3.6
Water (see 6.7)	500	4.4.3.11"

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Para 3.1.5 Properties. delete the Table appearing in this paragraph in its entirety and substitute the following Table IV.

Table IV

Property	Requirement	Test Paragraph
pH		
in water	≥ 2.0 to 7.0	4.4.3.12
in Methyl Formate (MF)	≥ 2.5 to 7.0	
Solubility in MF, %, min.	97.0	4.4.3.13
Hydrolysis/Dehdration		4.4.3.14
Methanol, % max at 200hrs	12	
Dimethyl ether, % max at 200 hrs	0.1	

Para. 4.4.2 Testing. Add the following to the end of the sentence. "At a Relative Humidity not to exceed 4% at ambient conditions".

Para. 4.4.2.1 Make the following changes to this paragraph:
1st Sentence: change "1 pound" to "100 g".
2nd Sentence: add the following statement: "or the samples from each batch that is combined into a shipping lot shall be subjected to the tests listed below in accordance with MIL-STD-105, Level II.
Table: Delete Table given and substitute the following Table V.

Table V

Inspection/Requirement	Defect Classification
Form & Color (see 3.1.1)	Major
X-ray Diffraction (see 3.1.2) or	Major
Infra-red Spectroscopy (see 3.1.3) (Alternate Method)	Major
Total Halogens (see 3.1.4)	Major
Water (see 3.1.4)	Major
pH (see 3.1.5)	Major
Solubility in MF (see 3.1.5)	Major
Hydrolysis/Dehydration (see 3.1.5)	Major

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Paragraphs 4.4.3.4, 4.4.3.5, 4.4.3.7, 4.4.3.8, 4.4.3.9, 4.4.3.10, and 4.4.3.15 - delete these paragraphs in their entirety.

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4.4.3.6 Halogen: delete in its entirety and substitute the following:

"4.4.3.6 Total Halogens. Determine the halogens present in the compound by use of a standard analytical technique for total halogens which has prior approval. (see 6.8).

Para. 4.4.3.11 Water. To end of sentence add "(see 6.7)."

Para. 4.4.2.12 pH. Change "4.4.2.12" to "4.4.3.12." Delete this paragraph in its entirety and substitute the following:

"Para. 4.4.3.12 pH. Prepare a 1 Molar solution of the $LiBF_4$ by dissolving 93.75g of the sample in a liter of boiled distilled water, having a pH between 6 and 7, and determine the pH of the resultant solution in accordance with method given in ASTM-E70. Smaller solutions may be prepared with respective corresponding weight volumes of material reduced accordingly. In addition, prepare a 2 Molar solution of the $LiBF_4$ by dissolving 187.5g of sample in a liter of Methyl Formate. Quantitatively transfer 100ml volumetric flask and dilute to the mark with boiled distilled water, having a pH between 6 and 7. Determine the pH of this solution in accordance with the method given in ASTM-E70. The pH measurement should be taken immediately following preparation of the solution, with readings taken between 1 and 3 minutes following preparation."

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Para. 4.4.3.13 Solubility in Methyl Formate. At the end of the 1st sentence add the following: "Smaller solutions may be prepared with respective corresponding weights/volumes of material reduced accordingly:"

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Add a new paragraph.

Para 6.7 If the lot of lithium tetrafluoroborate fails to meet the water requirement specified in 3.1.4 the lot of lithium tetrafluoroborate may be reprocessed by vacuum baking at 80°C ± 5°C and then retested in accordance with 4.3.4.11."

Add a new paragraph.

"Para 6.8 - the method used to determine the requirement should be submitted to DRDAR-QAR-R thru the Contracting Officer for prior approval.

"The margins of this amendment are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

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